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**Politics, Identity, and Ontological Security
in Bosnia and Herzegovina**

Master's thesis

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Prague, April 28, 2026

Zachary Alan Stuck

Abstract

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Abstrakt

Nutnou součástí práce je anotace, která shrnuje význam práce a výsledky v ní dosažené. Anotace práce by neměla být delší než 200 slov a píše se v jazyce práce (tj. česky, slovensky či anglicky) a v překladu (tj. u anglicky psané práce česky či slovensky, u česky či slovensky psané práce anglicky). Anotace práce by neměla být delší než 200 slov a píše se v jazyce práce (tj. česky, slovensky či anglicky) a v překladu (tj. u anglicky psané práce česky či slovensky, u česky či slovensky psané práce anglicky). V abstraktu by se nemělo citovat.

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Acronyms

BiH	Republic of Bosnia and Herzegovina
EU	European Union
FBiH	Federation of Bosnia and Herzegovina
HDZ BiH	Hrvatska demokratska zajednica (Croatian Democratic Union)
IR	International Relations
MP	member of parliament
NATO	North Atlantic Treaty Organization
OS	Ontological Security
SDA	Stranka demokratske akcije (Party of Democratic Action)
SNSD	Savez nezavisnih socijaldemokrata (Alliance of Independent Social Democrats)
SDP BiH	Socijaldemokratska partija (Social Democratic Party)
SDS	Srpska demokratska stranka (Serb Democratic Party)
RS	Republika Srpska
NLP	Natural Language Processing
WTO	World Trade Organization

Chapter 1

Introduction

This document serves two purposes. First, it is a template and example for a master's thesis. Second, the text in all sections contains some useful information on structuring and writing your thesis.

The introduction should consist of three parts (as paragraphs, not to be structured into multiple headings): The first part deals with the background of the work and describes the field of research. It should also elaborate on the general problem statement and the relevance. The second part should describe the focus of the thesis, typically the paragraph starts with a phrase like “The objective of this thesis is” The last part should describe the structure of the thesis, for instance in the following manner. The thesis is structured as follows: Chapter 2 cites some formal requirements of the faculty, Chapter 3 gives some hints on basic formatting features and covers also acronyms, figures, boxes and tables. Chapter 4 gives a recommendation on the usage of hyphens in English language in $\text{\LaTeX}$ and explains how to use the `itemize` and `quote` environments and shows a few `enumerate`-based environments. Chapter 5 presents a checklist of common mistakes to avoid. Chapter 6 contains numerous hints. Chapter 7 summarizes our findings.

H_1 : There are universal topics and concepts in political speech among the political elite in Republic of Bosnia and Herzegovina (BiH), regardless of party, that shape the Bosnian state identity.

H_2 : BiH seeks ontological security

H_3 : Ontological security shapes the political discourse among politicians.

Chapter 2

Theoretical Framework

This thesis conducts analysis under the constructivist paradigm and uses Ontological Security (OS) theory as the understanding for analysis of political speech acts by politicians in BiH. Using this framework allows for a nuanced conceptualization of how political speech acts shape and are shaped by national identities, state actions, ethnic tensions, and other societal factors of state collectives.

2.1 Social Constructivism

Social constructivism provides an International Relations (IR) framework that is largely based on the concept of intersubjectivity—shared meaning and understanding—in society and how various actors influence these meanings and understandings. The primary assumption that constitutes this intersubjectivity is “that people act toward objects, including other actors, on the basis of the meanings that the objects have for them” (Wendt 1992, pg. 396-397). Furthermore, Wendt (1992, pg. 404-405) highlights that actions taken before, during, and after an interaction between actors shapes the meaning of said interaction and how it is understood between them. Despite the relative simplicity of these concepts, they are essential to the constructivist paradigm and allow for the collectivization of meanings and understandings. Building upon these fundamental assumptions, actors and groups of actors are able to create complex societies, interactions, and institutions that constitute modern society, particularly that of governments, states, and the international global order (Wendt 1992, pg. 401-402).

2.1.1 Identity

Identity is fundamental to actors or groups of actors, but they are not limited to one particular identity, but rather multiple identities that are situation dependent. As Wendt (1992, pg. 397) explains, “Actors acquire identities—relatively stable, role-specific understandings and expectations about self—by participating in such collective meanings.” The primary parts of identity, therefore, are an understanding of self and collective meaning under certain circumstances. This can be seen in BiH when discussing *Bosnian* identity and ones *ethnic* identity. Actors often identify as *Bosnian* when interacting with outsiders, but they identify as a particular *ethnic* identity when interacting with other *Bosnians* or members of their collective *ethnic* identities. Contextuality determines many aspects of identity due to its intersubjectivity, and this allows identities to change as context may also change (Wendt 1992, pg. 397-398).

Due to the reliance on roles, identities often overlap, such as someone being both a father and a manager. These two identities are not mutually exclusive, and may be beneficial or detrimental to each other under certain circumstances, but they are both reflections of self expectations and self understanding of one’s roles (Wendt 1992, pg. 397). When applying this to collective identities, such as national identities, ethnic identities, etc., there is a certain role that members of the collective assign to that identity, but also to other identities of a similar collective level (Weldes 1996, pg. 280). These identities require other collective identities of a similar level—nations require other nations—because these identities are “mutually constitutive” (Giddens 1979, as cited in Wendt 1992, pg. 399). Not only are identities formed by internal identification of *self*, but also from external identification that is interpreted by the actor or group of actors who belong to that identity (Todorova 2009, pg. 10-11).

On a national level, identities are formed by the collectivization of people that understand some belonging to a nation, but the identities themselves are shaped by the actions of statesmen (Weldes 1996, pg. 281). National identity is still a collective understanding by the members of the nation, but the norms that form the institution of government provide statesmen with the agency to act and interpret meaning on behalf of the nation (Weldes 1996, pg. 281). This agency provides the grounding for understanding the function that political speech acts in the formation, change, or dissolution of national identities.

2.1.2 Interests

Following the formation of identities, actors form interests that correspond to one or more identities (Wendt 1992, pg. 398). These interests are highly dependent on situations and are motivated by anticipated outcomes of each situation (Wendt 1992, pg. 398), but these interests also serve to reproduce the understood meaning of an actor's identity (Mitzen 2006, pg. 347). These interests come both from the actor's understood meaning of identity, but also as a societal understanding that has become institutionalized, as all social interactions and understanding are inherently intersubjective (Wendt 1992, pg. 398-399; Weldes 1996, pg. 280-281).

Similar to national identities, national interests are also primarily formed by statesmen in alignment with their understanding of the national identity, the situation, and potentially their personal interests (Weldes 1996, pg. 280-282). Weldes (1996, pg. 280) critiqued Wendt's (1992) interpretation of states as unitary actors, and posed that "the meanings which objects, events and actions have for 'states' are necessarily the meanings they have for those individuals who act in the name of the state." This interpretation does not mean that individual citizens of a state or nation do not impact national interests—citizens elect these statesmen in democracies, and protest provide a voice to contest actions made by statesmen—but it puts the primary formation of national interests in statesmen's hands.

2.2 Ontological Security

The use of OS in IR research has increased significantly since it was introduced to the field by Kinnvall (2004) and Mitzen (2006), who extended the original application from sociology (Giddens 1991) to states and international actors. OS relies on actors maintaining consistent identities, routines, trust, and other cognitive and social structures, which are often considered making up the *security of being* (Kinnvall & Mitzen 2020, pg. 240). Giddens (1991) focused on the OS of individuals, particularly the experiences of individuals and how they shape one's understanding of themselves. As Mitzen (2006, pg. 352) poses, a similar approach to analyzing *the security of being* can be applied to states as collective units formed by societal interactions and understanding. OS in IR expands upon the paradigm of social constructivism by applying Giddens's (1991) understanding of routines, trust, self to the state level (Mitzen 2006;

Kinnvall 2004). This expansion constructed an analytical framework that provides an understanding of OS and insecurity, as well as the ability of states to form interests and take actions with an inherent interest to maintain OS and reduce uncertainty (Mitzen 2006; Kinnvall 2004).

2.2.1 Identity

Similar to traditional social constructivism, identities motivate actors' interests, actions, and understandings, but OS expands the role of identity and poses that actors are motivated to maintain stable identities to feel ontologically secure (Mitzen 2006, pg. 344-345). The literature on OS also expands upon the formation of identities, particularly group identities by highlighting how humans understand *self* and *self-differentiation* (Kinnvall 2004, pg. 749-750) Kinnvall (2004, pg. 749-752) poses that self-identification inherently requires a *subject other*, and he debates the difference between categorization, both internally and externally, and membership regarding the role of societal groups in identity formation. Furthermore, Kinnvall (2004, pg. 751-752) combines securitization with the subjectivity of *other* to describe how both an actor's self-identity and subjective understanding of the *other* are securitized to maintain a cohesive interpretation of society and the environment.

The way that identifies are essential to their securitization and to the securitization of the self. In Kinnvall's (2004) construction of OS, the concept of *abjection*—"unconscious self is neither an object nor a subject" (Kinnvall 2004, pg. 753)—is a way to reduce uncertainty and increase OS, both for individuals and collectives (Kinnvall 2004, pg. 753-754). This unconscious state of abjection is a primary method for mitigating change, but it leads to the creation of collective identity "when the familiar 'stranger' is suddenly recognized as a threat" (Kinnvall 2004, pg. 753). This can be seen clearly in BiH following the collapse of Yugoslavia and reemergence of ethnic identities that were weaponized to insight the ethnic conflicts in the 1990s. The dichotomy between self and other that leads to the creation of or membership to identities is particularly evident and destructive when the differences between self and other become ambiguous (Norton 1988, as cited in Kinnvall 2004, pg. 753).

This rather psychological approach to the formation of identity highlights the importance of maintaining OS and reducing the ambiguities of the self. When actors are unsure of who they are or what is to come, they often turn to membership or creation of identities driven by fear or hatred of the *other*

rather than the simple dichotomous classification between self and “stranger other” (Kinnvall 2004, pg.754 755). This formation of enemies is what leads to racism, anti-Semitism, ethnic conflicts, genocides, and conflict in general. This is not to say that ontological insecurity guarantees the formation of ‘enemies,’ but it does increase the chances of these classifications forming.

2.2.2 Routines & Trust

OS’s goal of mitigating uncertainty and maintaining a consistent understanding of self and being relies largely on routinization and trust (Mitzen 2006, pg. 346). An actor is never able to account for all possible outcomes in a situation, which is why Giddens (1991, pg. 36-37) poses the concept of “bracketing” as a method that actors use to mitigate uncertainty, and this is formed by a trust in certain “acceptable” responses or reactions that are understood within a society. Bracketing includes virtually any form of knowledge, whether it is understood by society or just the actor, such as scientific facts, societal expectation, or identity, and a disruption to what is understood leads to ontological insecurity (Giddens 1991, pg. 37-38; Mitzen 2006, pg. 346-347). Mitzen (2006, pg. 346) coins the collection of these “bracketings” as “an actor’s *basic trust system*”, and this trust system is largely influenced by the various identities that an actor holds, providing the framework for how and why an actor has certain understood truths (Giddens 1991, pg. 37-40).

This *basic trust system* is the result of routinization and the unconscious decisions that result from routines (Mitzen 2006, pg. 346-347). The mitigation of cognitive choice resultant of routines forms the bracketings and trust systems used by actors to limit uncertainty, justify identities, and propagate OS (Mitzen 2006). Routines also serve as a form of risk mitigation, both physical and ontological, as actors must always balance the possibility of risk through *risk profiling* (Giddens 1991, pg. 127-130). Despite this, attachment to routines may actively disregard risk out of fear or risk of altering or losing an identity (Mitzen 2006, pg. 347). As Mitzen (2006, pg. 350-351) claims, a “rigid basic trust system”, heavily reliant on routines to mitigate disruption in self-understanding and identity, opens an actor to ontological insecurity due to the lack of cognitive distance from the routines themselves, which leads actors with “rigid basic trust systems” to act irrationally in order to preserve routine. This irrationality caused by routines and trust systems can be used to explain a variety of decisions

that may detrimentally impact actors in order to maintain routines and the status quo.

2.2.3 Narratives & Collective Memory

Narratives and collective memory are essential to the creation of national and ethnic identities, and they provide a framework for the collective understanding required of collective and institutionalized identities. States often control narratives through direct and indirect methods of discourse and action, political speech acts, memorialization of events, public holidays, legislature, and foreign policy (Subotić 2015, pg. 615-617). National and ethnic narratives are constructed and maintained by “narrative entrepreneurs”—political and intellectual elites, media, and education systems—and solidified through daily engagement with these narratives by the public (Subotić 2015, pg. 615). Narratives act as a form of routine, as they provide meaning to state actions and form “autobiographies” for states that shape trust and collective memory (Subotić 2015, pg. 614). When narratives are seen as “rhetorical commonplace,” they are difficult to dispute and become essential to the understanding of self, both of the state and its citizens (Subotić 2015, pg. 615). This solidification of narratives “brackets” political debates and requires politicians to frame policy and discourse in alignment with narratives (Subotić 2015, pg. 615-616).

2.3 Ontological Security in BiH

BiH has a unique position regarding OS as a state primarily constituent of three ethnicities, Muslim Bosnjaks, Catholic Croats, and Orthodox Serbs (Bringa 2000, pg. 85). Following the collapse of the Socialist Federal Republic of Yugoslavia in, BiH declared its independence in 1992, and was recognized by the European Community (David 2019, pg. 216). Following independence, BiH descended into conflict between the three ethnicities, Orthodox Serbs, Catholic Croats, and Muslim Bosniaks from April 1992 to October 1995, leading to immense damages, casualties, and displacement across BiH (David 2019, pg. 216). The conflict officially ended in December 1995 with the Dayton peace agreement, which recognized BiH as an independent nation-state and defined the constitution that is still used in BiH (David 2019, pg. 216). The constitution provides a unique institutional framework that divides almost all functions of the government between the three major ethnicities (OSCE Mission

to Bosnia and Herzegovina 1995). As well dividing the institutions among the three ethnicities, two separate entities govern territory in BiH—Republika Srpska (RS) which governs much of the Serbian ethnic areas in the north and west of BiH, and Federation of Bosnia and Herzegovina (FBiH) which governs much of the Croat and Bosniak ethnic areas in the west, south and center of BiH (OSCE Mission to Bosnia and Herzegovina 1995). Beyond the institutions of BiH, three of the four major parties are also ethnically aligned with Alliance of Independent Social Democrats (SNSD) being Serb aligned, Croatian Democratic Union (HDZ BiH) being Croat aligned, and Party of Democratic Action (SDA) being Bosniak aligned (Parlamentarna skupština 2026). Not only are these parties ethnically aligned, but HDZ BiH is a sister party to HDZ in Croatia, and Social Democratic Party (SDP BiH), which is the successor to the League of Communists of Yugoslavia, has sister parties in Croatia, Serbia, and North Macedonia. The complicated history and institutionalization of ethnic tensions leads to a unique case for OS theory to address.

As highlighted above, OS theory stresses that states seek to maintain stability in identity, routine, and trust through various means, but this becomes complicated in states like BiH that have institutionalized divisions between ethnicities (See Section 2.2). The state seeks to maintain the *Bosnian* identity, which encompasses all citizens and is what is projected externally to other states, but the various ethnicities also seek to maintain their respective ethnic identities, which may not always align with the state's projected identity. This complexity results in the opportunity to analyze how BiH operationalizes OS and the extent to which it is ontologically secure or insecure as a state.

Identity is largely sustained through narratives, which often become engrained in collective memory as “rhetorical commonplace,” but actors are able to leverage these narratives through the *activation* and *deactivation* of narratives or parts of narratives, particularly in regard to political speech and policy (Subotić 2015, pg. 615-616). This can be seen often in politics, and it is crucial for political discourse, debate, and policy. This can particularly be seen regarding events in the collective memory of BiH, such as the war in the 1990s or World War II, leading to the possibility for collective memorialization of tragedy across all Bosnians or the weaponization of these memories for political gains (David 2019). The variation in narratives or parts of narratives have a major impact on the meaning of certain events, speech acts, and policy, and depending on how they are interpreted can support or hurt the understanding of state identity.

Part of Bosnian identity relies on

Political speech acts are a primary way of understanding the operationalization of OS theory at a state level because politicians have been given agency to speak for the social group that constitutes a state, and their speeches and actions are publicized to both internal and external actors (Weldes 1996, pg. 280-281). It is ultimately politicians who speak and act on behalf of the state, and the interactions between politicians largely take place through parliamentary debate. It is the aim of this thesis to analyze how Bosnian politicians' speech acts in parliament shape the state's identity, routines, and trust, and if this aligns with OS theory despite the complexity of Bosnian society and institutions.

Chapter 3

History & Context

Chapter 4

Methodological Framework

Ontological security relies on analysis of discourse and socio-cognitive aspects of society to determine how groups, primarily states in IR, form and maintain stable identities, routines, and trust (Mitzen 2006; Kinnvall 2004; Subotić 2015).

4.1 Methodology

4.2 Data Collection & Preparation

The data used for analysis comes from the BiHCorp, which is a dataset that contains over 127 thousand speeches from transcriptions of debates in the Parliamentary Assembly of Bosnia and Herzegovina between 1998–2018 (Mochtak *et al.* 2022). The relevant variables for this analysis are *house*, *date*, *fullname*, *party*, and *speech* (Mochtak *et al.* 2022). The data was then constrained to a selection between 2010–2018 due to hardware limitations and scope. This working dataset was then machine translated using the Helsinki-NLP/opus-mt-sla-en model for Slavic languages to English (Tiedemann & Thottingal 2020). This model was chosen as an open source alternative to services such as Google Translate or DeepL due to the quantity of data and the model’s comparable performance in South Slavic languages (Tiedemann & Thottingal 2020). Machine translation was necessary to ensure strong performance across various machine classification models, which are primarily trained on English language corpora, and due to the researcher’s limited command of the Bosnian language. The use of machine translation allows for analysis on multilingual corpora or corpora outside a researcher’s spoken languages while maintaining the majority of linguistic context and meaning (Mate *et al.* 2023, pg. 7-8).

Following the preliminary data preparation and translation, the corpus was further refined to minimize the number of procedural speeches. Procedural speeches entail voting procedures, recess announcements, and other technical aspects or parliamentary proceedings. This was classified with the assistance of the ParlaCAP model Kuzman Pungeršek & Ljubešić (2025), which is an XLM-RoBERTa-large based model that was finetuned using proceedings from 29 European countries, to classify particular topics in parliamentary debate, although only the “other” classification was used for constraining the data to ontologically important speech. The final preparation step was segmenting each speech to fit the context window of the model for training and classification. The RoBERTa-large model has a context window of 512 tokens, and every speech was segmented to a maximum of 512 tokens for manual and machine aided classification. Table 4.1 shows the number of samples throughout this collection and preparation phase. All scripts used throughout the analysis can be found in Appendix B.

Table 4.1: Total sample size throughout data preparation

Stage	Sample size
Initial	127,713
2010-2018	55,943
Filtered	26,353
Segmented	120,783

4.3 Model Training

This analysis required the training of a fine-tuned **RoBERTa-base** model, which was used for assigning relevant classifications to parliamentary debates. This model was chosen due to its proven effectiveness at similar tasks, and the use of a pre-trained model minimizes extensive training, corpus collection, and computing resources required to create effective multi-classification models (Burst *et al.* 2024; Timoneda & Vera 2025, pg. 2-3) All use and training of models utilized the *transformers* architecture, which is a standard pipeline for Natural Language Processing (NLP) (Wolf *et al.* 2020, pg. 38-40).

4.3.1 Text Classification & Active Learning

A codebook was developed to capture discursive elements that have been deemed relevant to the formation of ontological (in)security, while limiting its size for increased model precision (Mitzen 2006, pg. 344-348; Kinnvall 2004, pg. 757-759; R. Abas *et al.* 2022, pg. 2945-2947). This codebook was then used for the classification of the initial training dataset and all subsequent amendments to it. This codebook includes aspects related to how actors identify themselves in their speech; the use of certain narrative devices like historical contextualization, moral agency, and the use of collective memory; and who or what the actor “others” (Van Dijk 1993, pg. 259-265).

An active learning approach was taken to increase the rate of model training and to provide training examples for classifications that are statistically rare in the dataset. Active learning involves manually classifying data in small batches, training the model, and then evaluating which data should be classified next based on the model’s uncertainty (Vacareanu *et al.* 2024, pg. 322-323). There are various approaches to active learning, but the two criteria used were entropy (uncertainty) and rarity. Entropy represents the model’s certainty of a prediction, and a high entropy represents a high level uncertainty (Vacareanu *et al.* 2024, pg. 323).

$$H(x) = - \sum_{i=1}^n P(y_i|x) \log_2 P(y_i|x) \quad (4.1)$$

Equation 4.1 was used in for all but 2 training rounds, and 50–70% of each round drew from the pool of data that had the highest entropy. Although methods such as focal loss provide more cohesive statistics for rare classifications, the selection of classifications was done manually to address fluctuations in performance of particular classes due to small sample size. At the beginning of active learning, these data points accounted for 30% of each round, but the last 100 samples were exclusively rare classifications to address imperfections in the training set. In each training round, the total dataset of manually classified texts was split into training and validations sets using “multi-label iterative stratification,” as it accounts for the imbalance in the dataset and provides enough validation samples from each category to accurately show the metrics for the model (Sechidis *et al.* 2011, pg. 146-148). Table 4.2 shows the metrics of the model at every training stage, which reflect the model’s ability to accurately classify political texts from the Parliamentary Assembly of Bosnia and Herzegovina in accordance with the

codebook created for this analysis.

Table 4.2: Training performance by iteration

Sample Size	Validation Loss	F_1 Macro	F_1 Micro	Hamming Loss
311 samples	0.351	0.251	0.519	0.152
400 samples	0.316	0.340	0.632	0.130
500 samples	0.236	0.583	0.776	0.088
600 samples	0.144	0.870	0.915	0.036
650 samples	0.112	0.887	0.934	0.028
700 samples	0.108	0.895	0.939	0.026

4.3.2 Model Performance

The final version of the model shows strong results using the final 700 sample training and validation set. The general results seen in table 4.2 and the results of the individual classifications in table D.1 are very strong, showing that the model is 89.5% accurate overall, over 90–95% accurate 11 of the 17 classifications, and only 2 classification have 66% accuracy. The primary indicators of the model’s performance is the F_1 score (see equation 4.2), which accounts for the model’s precision and recall, but the model also maintains extremely high precision (see equations B.1 and B.2) (R. Abas *et al.* 2022). This is indicative of a model that primarily classifies accurately and avoids inaccurate predictions, as seen by the high precision but lower recall values.

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.2)$$

The lower performance the `COND_CORRUPT` or `OTHER_ABSTRACT` categories is likely due to the limited sample sizes available for training. Beyond explicit mentions of “corruption,” the model struggles to identify the nuanced linguistic markers of the `COND_CORRUPT` class. As shown in Table D.1, this category achieves perfect precision but only 50% recall, illustrating a systemic “hesitance” in the model’s predictions. Conversely, the `PROCEDURAL` class, characterized by highly predictable formulas like speaker introductions, proposed legislation, and other technical language, yields an F_1 score of 1.00, indicating near-perfect identification.

As well as the model’s strong metrics, training of encoder based models for political science has been increasingly used and validated, particularly with datasets of similar size (Laurer *et al.* 2024, pg. 85-87). The accuracy of this

model is comparable to those discussed by Laurer *et al.* (2024, pg. 85), and it allows analysis of significantly larger datasets while maintaining high accuracy. When comparing the accuracy of a transformers based model to measures such as Cohen’s kappa, used for inter-coder reliability, the model performs with comparable results to classification by multiple humans (Coleman *et al.* 2024, pg. 140-141). Coleman *et al.* (2024, pg. 141) explain that a Cohen’s kappa of 0.81 or higher signifies almost perfect agreement between coders, and a model’s F_1 score is comparable to a kappa. Due to the resources available to this analysis, only one coder was the source for classification data, meaning that inconsistencies in the training data, or disagreement between the coder and the model can also explain the F_1 scores.

4.4 Machine Assisted Analysis

All models used, including the model trained for this analysis are available for use via HuggingFace, a community hub for sharing models, and the HuggingFace pipeline was implemented for all task where suitable using the *Transformers* library (Wolf *et al.* 2020, pg. 38-39). To minimize storage space, all results were stored as “Apache Parquet” files, and the analysis utilized the HuggingFace Dataset objects for efficient batching at scale (Apache Parquet Project 2025; Wolf *et al.* 2020, pg. 38-39).

After model training was complete, the model was applied to the entire dataset with a confidence threshold of 0.50 (Pungeršek *et al.* 2026, pg. 5). This data could then be further used throughout the analysis to identify ontologically relevant trends and segments across parliamentary speeches.

As well as the custom model for this analysis, two other transformers based models were used as well. The first is **ParlaSent**, which is a fine-tuned model used for sentiment analysis of parliamentary debate, which is based on **xlm-r-parla** and trained on 18 thousand parliamentary debates across 7 languages (Mochtak *et al.* 2025). This model was partially trained using data from the Parliamentary Assembly of Bosnia and Herzegovina, further validating the use of this model and others on this dataset (Mochtak *et al.* 2025, pg. 4-5). The model returns sentiment on a 0–5 scale, which is used in later statistical analysis comparing certain topics and ontologically relevant categories to determine overlap and significance. Although the model is able to analyze sentiment in the Bosnian language, the sentiment was applied to

the translated segments to maintain consistency across the entire methodology (Mochtak *et al.* 2025, pg. 6).

The last model that was used for this analysis was **BERTopic**, which is a model trained to cluster texts based on their embeddings, similar to traditional TF-IDF (Grootendorst 2022, pg. 2). This is used to find common topics across the parliamentary debates, and allows for understanding how certain topics coincide with different classifications and sentiments. By creating these topics based on similarities specifically in the texts, the analysis is able to determine the primary discussion points driving ontological (in)security. **BERTopic** has been used in a variety of different application across political science, and is a recognized standard for topic analysis on large datasets (). The model was only applied to texts that were classified at substantial to the analysis, as applying this to all texts would largely skew the results toward procedural topics due to the granularity of **BERTopic** ($N = 55,976$). To ensure replicability, UMAP and HDMSCAN parameters were set (see Appendix C), which control dimension reduction and clustering respectively (Grootendorst 2022, pg. 1-3; McInnes *et al.* 2020, pg. 1-3; McInnes & Healy 2017, pg. 1-2) As well as controlling these parameters, the model also utilized **KeyBERTInspired**, which is an additional model that is used to increase the readability of the topics found by the base **BERTopic** model (Grootendorst 2022)

By combining these 3 models, this analysis can draw upon the topics discussed, how certain discursive elements are used, and how positively or negatively each of these aspects are portrayed. These metrics allow for comprehensive empirical analysis that also informs the broader contextual analysis in the following chapters.

Chapter 5

Empirical Analysis

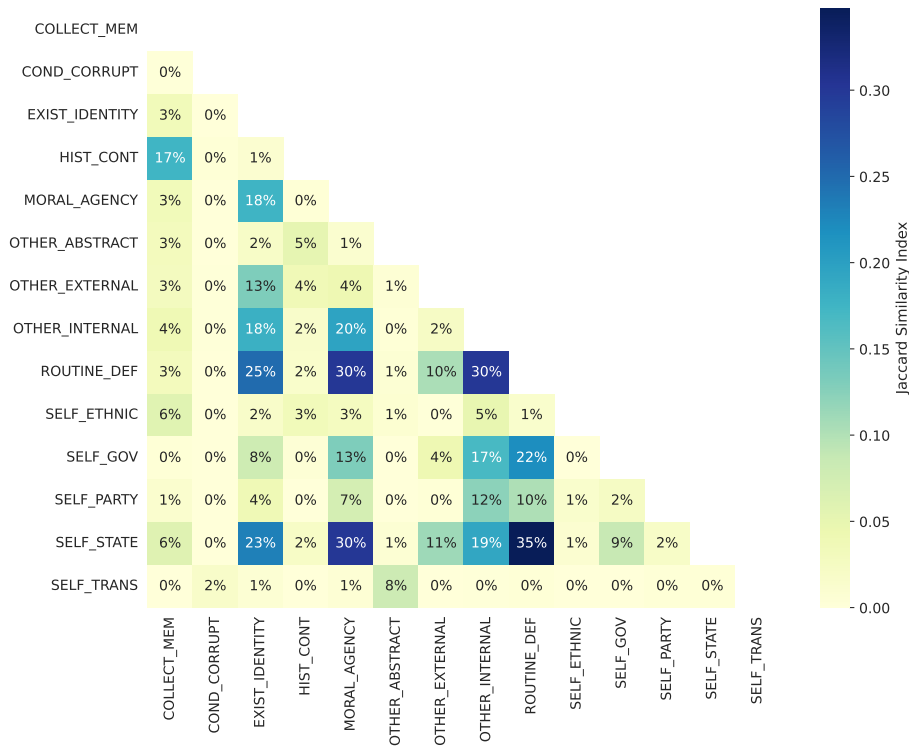
Following the classification, sentiment analysis, and topic modelling, a variety of statistical analysis can be applied to address the roll of political speech in identity, routine, and trust. For the purpose of analysis, only speech segments that were classified as `SUBSTANTIVE_IDENTITY` are included in the statistical analysis ($N = 55,976$) (see Table D.2). Table 5.1 show the total number of segments by classification and the relative frequency across the substantive segments, showing that the most common classification revolve around the state and another actor within the state, while direct condemnation of corruption or broad identification of self were rarely classified.

Table 5.1: Total count and relative frequency of substantive segments by classification

Classification	Count	Relative Frequency
<code>SELF_STATE</code>	21,632	38.65%
<code>OTHER_INTERNAL</code>	21,448	38.32%
<code>ROUTINE_DEF</code>	19,152	34.21%
<code>SELF_GOV</code>	18,377	32.83%
<code>MORAL_AGENCY</code>	12,254	21.89%
<code>EXIST_IDENTITY</code>	8,238	14.72%
<code>OTHER_EXTERNAL</code>	3,400	6.07%
<code>SELF_PARTY</code>	3,353	5.99%
<code>COLLECT_MEM</code>	1,478	2.64%
<code>SELF_ETHNIC</code>	1,123	2.01%
<code>HIST_CONT</code>	694	1.24%
<code>OTHER_ABSTRACT</code>	195	0.35%
<code>SELF_TRANS</code>	75	0.13%
<code>COND_CORRUPT</code>	50	0.09%

This correlates with the qualitative understanding of these texts, but the model’s ability to classify these minor classifications may also be a result of limited samples in the training data set. Due to speakers being able to leverage multiple identifications of self and other, as well as multiple rhetorical devices, many classifications have an overlap. Figure 5.1 shows the Jaccard similarity index (Equation B.5) for every classification, and a few groups stand out. The largest overlap is the identification with the state and routine definition, signifying these two are commonly used together in speech. Similarly, routine definition is also common with internal othering and moral agency. These overlaps provide insight into the common patterns used in political debate in BiH, and these patterns show significant effects on how political speech is framed and the sentiment used.

Figure 5.1: Overlap of classifications shown with Jaccard Similarity index



5.1 Archetype Clustering

To address this overlap for further use in statistical analysis, these classifications were clustered into five rhetorical archetypes. The method used to create these archetypes was k-means clustering, a commonly used algorithm for grouping

data into categories based on the mean value of certain attributes (Ikotun *et al.* 2023, pg. 179-180; Senn *et al.* 2025, pg. 7-9). Although this is primarily used in data science or natural sciences, Senn *et al.* (2025) utilized this method to identify different communication styles used by politicians during the Covid-19 pandemic. These clusters not only provide insight into the structure of debates in BiH, but also who uses certain structures.

The clustering of the classifications was computed in python with the **prince** and **scikit-learn** packages, and the outcomes can be found in table D.3 (Halford; Pedregosa *et al.* 2011). This clustering was computed on a subset of the data that consisted of only political parties with over 2,000 segments present in the data to prevent outliers from skewing the algorithm, and to maintain the same sample size for analysis regarding the effect of party membership on political speech. Furthermore, 5 clusters were chosen to understand the larger patterns used in political speech, and these broader archetypes capture the types of self and other identities and the rhetorical devices used (Senn *et al.* 2025, pg. 9). These 5 clusters were then assigned to the texts according to their classifications and the classifications belonging to these clusters. The total count of segments belonging to clusters can be seen in table 5.2, which shows a high use of internal governance and institutional othering.

Table 5.2: Total count of segments per rhetorical archetype

Rhetorical Archetype	Count
Internal Governance	19500
Institutional Othering	7816
Existential Statehood	7629
Civic/Moral Statehood	7068
Historical/Memory Statehood	1592
<i>Note: N = 43,605</i>	

These archetypes are generalized for analysis, but examples from each one show the actors, tone, and sentiment commonly associated with each archetype. The most common archetype is *internal governance*, which predominantly deals with government policies and remains largely diplomatic in daily proceeding. It predominantly uses the government and its institutions as the “self” identification of the actor or group, and it identifies the “other” more politely than archetypes like *institutional othering*. An example of this can be found in a segment of debate by Ilija Filipović (HDZ BiH) on BiH’s intelligence and security policy (Mochtak *et al.* 2022; Translated via OPUS-MT):

The thing needs to be resolved, if this is the question, this is really a matter of an interstate agreement. If we're looking at such a thing, then it really isn't the place here, but maybe these things that you've been talking about can be brought into line with legal and international legal practice, international legal standards and domestic legislation to put it in place on the Joint Commission instead...

He uses the term “we” to address his colleagues in the parliament, and avoids combative language or direct conflict with other parties or members of parliament. This archetype also largely aligns with topics directly related to policy rather than history, ethnic groups, or even the Bosnian people as a whole. This context regarding this archetype explains the significant frequency of it among the segments coded, and shows that political speech does maintain the decorum expected of parliament.

The second archetype, and one of particular interest is the *institutional othering*. This archetype is the second most common across the dataset analyzed, and it exemplifies the internal tensions that persist in BiH. This archetype often includes direct condemnation of politicians, political parties, institutions, or ethnicities with loaded language. The use of routine definition and moral agency is also common, and they are often used to strengthen the speaker's point and attack the “other.” Although this rhetoric can include “self” identification with political party or ethnic group, it usually includes some form of governmental identification regardless of any others. An example of this archetype's use can be found in a segment of Mladen Bosić's (Serb Democratic Party (SDS)) debate on the proposal of a law regarding the termination of laws (Mochtak *et al.* 2022; Translated via OPUS-MT):

Should it be said that these for the Court and the Prosecutor's Office are certainly second-rate victims of Serbs, and first-class victims are known. And all mothers, regardless of faith, nation and color, have the same pain for lost children. This is against all morals and such selectivity, which is easy to express in numbers, created in RS a conviction that it is not about justice but at the very least about the selective justice of the Court and the BiH Prosecutor's Office...

This segment includes many of the common elements found in this archetype, such as the direct mention of his personal ethnic group and the entity of RS, moral agency and condemnation of other member of parliaments (MPs) and

parties, and undermining of BiH's institutions. The use of "victims" can also be found in this and other archetypes, and there can be a variety of direct attacks on character, morals, or ethnicity that further weaponize and charge the language used. *Institutional othering* does not always include such charged language, because it also includes simple disagreements over policy that did not fit into *internal governance*, but its use is very common in throughout debate in BiH. The use of *institutional othering* also carries importance to OS, as internal othering often perpetuates a particular narrative that the actor is portraying, either of the state, party, or ethnic group (Subotić 2015, pg. 613-614; David 2019, pg. 223-224).

The next three archetypes are all related, but they vary in their framings and devices used. The first of these is *existential statehood*, which primarily uses the identification with the state and routines, while claiming that there is an existential threat to one of the actors identities. There are similarities to *institutional othering*, but rather claims that someone or something is a direct threat to the state, its identity, or a smaller subset. An example of this can be found in a segment of Drago Ljubičić's (SNSD) answers to another MP about Islamic radicalism (Mochtak *et al.* 2022; Translated via OPUS-MT):

These groups are not without organisation, they are, in my own opinion, they are not without financial support, and it is even possible to have political support in certain circles in the Federation of Bosnia and Herzegovina and, as such, are certainly a danger and a threat to security in BiH. Perhaps, gentlemen, they are also tied to the latest developments in the Federation of Bosnia and Herzegovina regarding meetings and posters for the transition of Orthodox and Catholics to Islam and who knows what it means after this...

This quote highlights a few key aspects of this archetype, such as something—radical Islamic groups—causing disruptions or threats to a FBiH and BiH. Similarly, it leverages the religious ties of each ethnic group and claims that Islamic groups are converting Christians, which is a threat to ethnic identities (Kinnvall 2004, pg. 759). This is one way MPs leverage this archetype, but another issue that is often brought up is BiH's entry into international organisations such as the European Union (EU), the World Trade Organization (WTO), and the North Atlantic Treaty Organization (NATO), and any actions that threaten this. This existentialism plays a large role in Bosnian politics, and is one of the primary ways that MPs weaponize the state's identity with particular framings.

Existential statehood also provides insight into the way OS is sought by MPs in the name of the state or other identities (Subotić 2015, pg. 614).

The second form of leveraging state identity comes as *civic/moral statehood*, which frames discourse in terms of civic duty, rights, and shared moral standards. This is typically used in more neutral or positive matters of obligations of the state, MPs, and citizens, and it often aligns with set frameworks, narratives, and routines in place. A segment of Hazim Rančić's (SDA) debate on public procurement policy highlights these characteristics well (Mochtak *et al.* 2022; Translated via OPUS-MT):

The first thing we insist on for ourselves, not for the sake of ourselves, or for the club [party], or both, but for the benefit of all citizens and all institutions involved in the public procurement process. Thus, changes and additions are aimed at facilitating and improving the process of public procurement. And, regarding the position of the clubs [parties], so I just want to say that the club [party] of Bosniaks has the same position as the club [party] of Croats in the House of the People of the Parliamentary Assembly of BiH...

Rančić highlights the obligations of the institutions of BiH and the MPs to act in the interest of all citizens and frames the issue as one that serves a civic duty. This form of *civic statehood* leverages the routines and identity of the state and reinforces them, largely in positive framings. Unlike *existential statehood*, this does not frame something as a threat to the established norms, but rather as a reinforcement and obligation to uphold them, further perpetuating OS (Subotić 2015, pg. 615-616; Mitzen 2006, pg. 251-253; Weldes 1996, pg. 276-277).

The final archetype, *historical/mnemonic statehood*, relies on collective memory and historical context to frame certain issues or topics. This form of discourse has been covered by (David 2019, pg. 216-219) in BiH, and this form of memory politics is common among the OS literature (Subotić 2015, pg. 614-615; Kinnvall 2004, pg. 755-757). Collective memory and historical context, particularly in a national context drive much of political and social discourse, even if not directly mentioned, and the use of this archetype perpetuates these memories (Kinnvall 2004, pg. 755-757).

5.2 Topic Analysis

Figure 5.2: Overlap

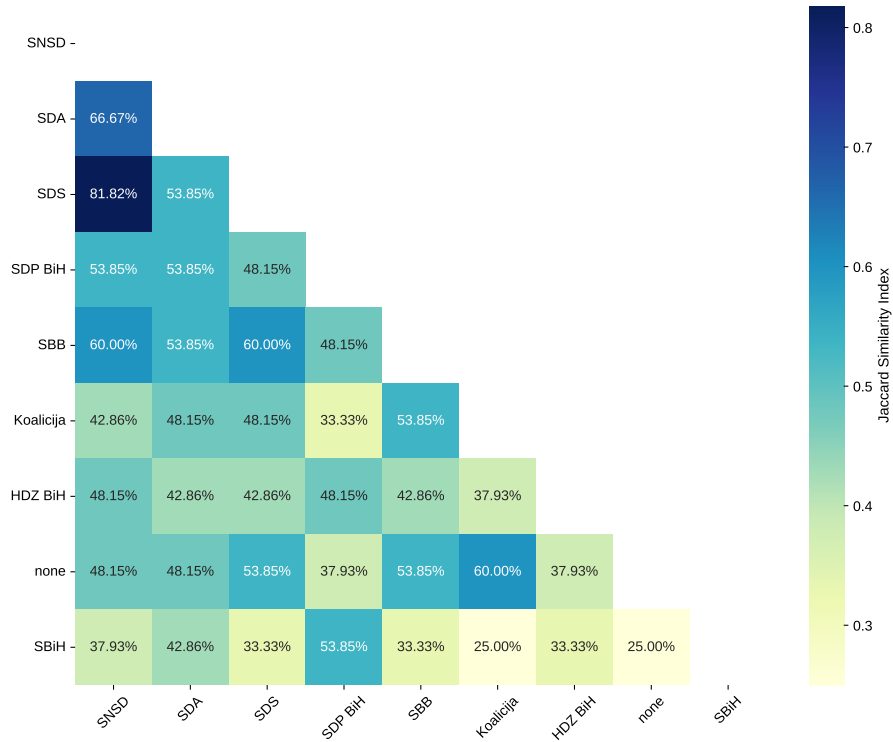


Table 5.3: Pearson Residuals: Party Identity vs. Rhetorical Archetypes

Party	Rhetorical Archetypes (Pearson Residuals)				
	Civic/Moral Statehood	Existential Statehood	Mnemonic Statehood	Institutional Othering	Internal Governance
HDZ BiH	0.099	-0.120	-4.713	-1.277	2.171
Koalicija	1.908	-3.643	-0.974	-5.812	5.088
SBB	5.812	-1.185	0.161	-4.972	0.344
SBiH	-0.685	-3.692	0.929	-2.887	4.284
SDA	4.199	-0.774	-2.297	-7.705	3.489
SDP BiH	1.390	3.198	5.461	-3.767	-2.012
SDS	-1.264	-3.816	-0.705	4.582	0.449
SNSD	-9.671	-0.247	2.440	15.685	-4.650
none	5.568	12.591	-2.923	-6.433	-6.320

Notes: $\chi^2 = 1047.52$, $p < 0.001$, Cramer's $V = 0.0775$.

Residuals $|r| > 1.96$ indicate significant association at $\alpha = 0.05$.

Figure 5.3: Mixed-Effects Model estimates for selected topics. Topics include the 10 most common, 5 most negative, 5 most positive, and 2 selected for comparison

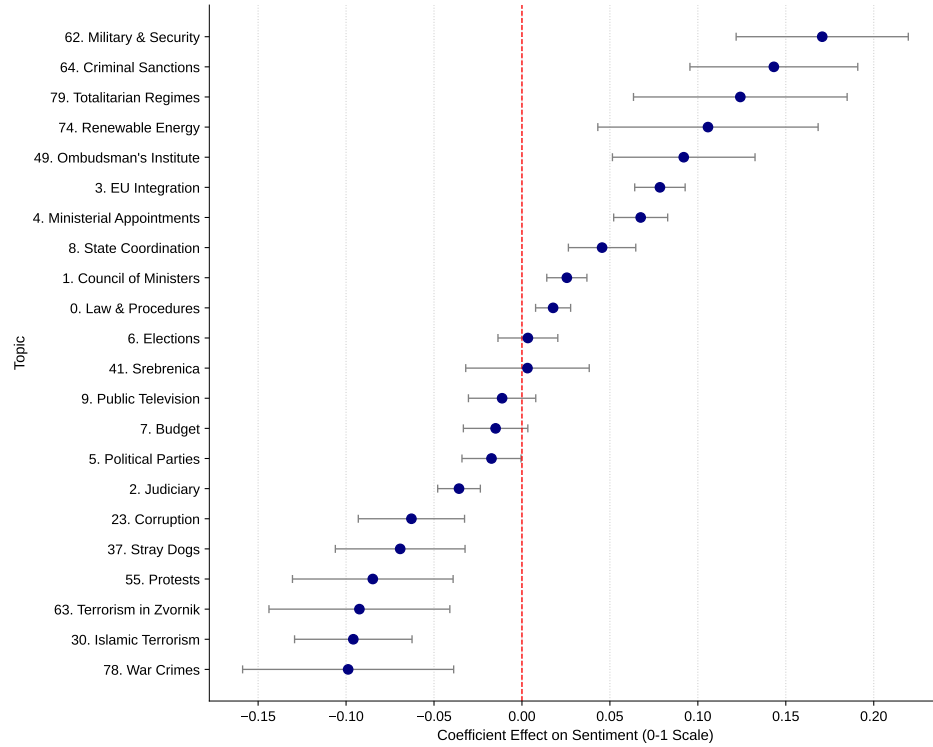


Table 5.4: Mixed Effects Linear Regression: Sentiment Drivers in the BH Parliament

Variable	Coef.	Std. Err.	z	P > z
<i>Fixed Effects (Intercept)</i>				
	0.434	0.026	16.63	0.000
Rhetorical Archetype (Ref: Civic Statehood)				
Existential Statehood	-0.141	0.025	-5.67	0.000
Institutional Othering	-0.176	0.021	-8.29	0.000
Internal Governance	-0.068	0.019	-3.50	0.000
Political Party (Ref: HDZ BiH)				
SNSD	-0.085	0.023	-3.71	0.000
SDS	-0.067	0.027	-2.50	0.012
Koalicija	-0.047	0.022	-2.14	0.032
Topic Highlights (Ref: State Coordination)				
Terrorism (Topic 30)	-0.208	0.039	-5.35	0.000
War Crimes (Topic 78)	-0.218	0.083	-2.63	0.009
Corruption (Topic 23)	-0.150	0.032	-4.71	0.000
Random Effects				
	Var.	SD		
Speaker (Fullname)	0.002	0.045		

Note: $N=13,695$. Interaction terms omitted for brevity; see Appendix B for full model.

Chapter 6

Contextual Analysis

The empirical analysis of speech patterns, self vs other, and general topics provides great insight into how Bosnian politicians speak, but this alone cannot explain how their speech shapes ontological (in)security. This chapter contextualizes and expands upon the empirical findings from the text classification, sentiment analysis, and topic modelling in order to greater understand what Bosnian politicians view as the most prominent issues in BiH and how they address them.

6.1 Dysfunction in BiH

A prominent theme across the 700 segments that were manually coded was the dysfunction of BiH's government especially regarding the three constituent peoples. This can be seen in quotes such as,

Here are the fundamental questions whether the will of one nation, another or third will be respected, and here is really the question—not the problem of the existence of three peoples, but in ... that we are not ready to admit it. And the third thing I want to do is we have a chance to get some lessons out of that 17-year-old post-Dayton fissure (Mochtak *et al.* 2022; Translated via OPUS-MT).

BiH

Chapter 7

Conclusion

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Appendix A

Data

This is optional: you may enclose a DVD to this thesis which contains empirical data and MatLab/R/Stata source codes. Even better so, you can create a special website for your project. Stating in your thesis that the data and source codes are available upon request is enough but please, have them prepared for such requests.

- Folder 1: Source codes
- Folder 2: Empirical data

Appendix B

Mathematical Definitions

$$\text{Precision} = \frac{TP}{TP + FP} \quad (\text{B.1})$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (\text{B.2})$$

$$\mathcal{L}_{val} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (\text{B.3})$$

$$\text{Hamming Loss} = \frac{1}{N \cdot L} \sum_{i=1}^N |y_i \oplus \hat{y}_i| \quad (\text{B.4})$$

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (\text{B.5})$$

Appendix C

Computational Workflow

Appendix D

Expanded Results

Table D.1: Model performance statistics including individual classification performance and overall performance

Classification	Precision	Recall	F1 Score	Support
COLLECT_MEM	1.0000	0.8235	0.9032	17
COND_CORRUPT	1.0000	0.5000	0.6667	6
EXIST_IDENTITY	0.8780	1.0000	0.9351	36
HIST_CONT	1.0000	0.7273	0.8421	11
MORAL_AGENCY	0.8500	1.0000	0.9189	51
OTHER_ABSTRACT	1.0000	0.5000	0.6667	14
OTHER_EXTERNAL	1.0000	0.7917	0.8837	24
OTHER_INTERNAL	0.9216	0.9792	0.9495	48
PROCEDURAL	1.0000	1.0000	1.0000	12
ROUTINE_DEF	0.9412	0.9796	0.9600	49
SELF_ETHNIC	1.0000	0.7000	0.8235	10
SELF_GOV	0.9630	0.8966	0.9286	29
SELF_PARTY	1.0000	0.8182	0.9000	11
SELF_STATE	0.9310	1.0000	0.9643	54
SELF_TRANS	1.0000	0.8000	0.8889	5
SUBSTANTIVE_IDENTITY	0.9890	1.0000	0.9945	90
SUBSTANTIVE_TECHNICAL	1.0000	0.9677	0.9836	31
Micro Avg	0.9451	0.9337	0.9394	498
Macro Avg	0.9690	0.8520	0.8947	498
Weighted Avg	0.9509	0.9337	0.9355	498
Samples Avg	0.9618	0.9517	0.9551	498

Table D.2: Total count of classifications

Classification	Count	Relative Frequency
SUBSTANTIVE_IDENTITY	55,976	46.34%
SUBSTANTIVE_TECHNICAL	39,544	32.74%
PROCEDURAL	23,708	19.63%
SELF_STATE	21,640	17.92%
OTHER_INTERNAL	21,461	17.77%
ROUTINE_DEF	19,156	15.86%
SELF_GOV	18,379	15.22%
MORAL_AGENCY	12,257	10.15%
EXIST_IDENTITY	8,241	6.82%
OTHER_EXTERNAL	3,406	2.82%
SELF_PARTY	3,354	2.78%
COLLECT_MEM	1,482	1.23%
SELF_ETHNIC	1,125	0.93%
HIST_CONT	696	0.58%
OTHER_ABSTRACT	196	0.16%
SELF_TRANS	75	0.06%
COND_CORRUPT	50	0.04%

Table D.4: Mixed Linear Model: Sentiment as a Function of Topic, Archetype, and Party

Variable	Coef.	Std.Err.	z	P> z
Intercept	0.434	0.026	16.627	0.000
<i>Main Effects: Topics</i>				
Topic 1 (Council of Ministers)	0.004	0.026	0.153	0.878
Topic 23 (Corruption/Crime)	−0.150	0.032	−4.709	0.000
Topic 2 (Prosecutor/Judicial)	−0.131	0.024	−5.393	0.000
Topic 30 (Terrorism/Islam)	−0.208	0.039	−5.349	0.000
Topic 37 (Stray Animals)	−0.159	0.037	−4.316	0.000
Topic 3 (EU Integration)	0.048	0.026	1.829	0.067
Topic 41 (Genocide/Srebrenica)	−0.080	0.050	−1.620	0.105
Topic 7 (Budget/Fiscal)	−0.131	0.029	−4.484	0.000
<i>Main Effects: Rhetorical Archetypes</i>				
Existential Statehood	−0.141	0.025	−5.672	0.000
Historical/Memory Statehood	−0.145	0.055	−2.649	0.008

Continued on next page

Table D.4 – continued from previous page

Variable	Coef.	Std.Err.	z	P> z
Institutional Othering	−0.176	0.021	−8.293	0.000
Internal Governance	−0.068	0.019	−3.496	0.000
<i>Control: Party Identity</i>				
Koalicija	−0.047	0.022	−2.143	0.032
SDS	−0.067	0.027	−2.500	0.012
SNSD	−0.085	0.023	−3.708	0.000
<i>Significant Interactions (Selection)</i>				
Topic 2 × Existential	0.126	0.033	3.780	0.000
Topic 41 × Existential	0.252	0.064	3.911	0.000
Topic 2 × Othering	0.126	0.029	4.301	0.000
Topic 7 × Othering	0.116	0.039	2.982	0.003
Group Variance (Speaker)	0.002	0.002		
Log-Likelihood	−284.392			
Observations	13 695			

Table D.3: Rhetorical Archetypes: Grouped Classification Probabilities
(N=55,976)

Classification Code	Probability
Archetype 0: Institutional Othering	
OTHER_INTERNAL	0.879
ROUTINE_DEF	0.728
SELF_GOV	0.434
MORAL_AGENCY	0.413
SELF_PARTY	0.315
Archetype 1: Existential Statehood (Weaponized)	
SELF_STATE	0.945
ROUTINE_DEF	0.848
MORAL_AGENCY	0.520
EXIST_IDENTITY	0.484
OTHER_INTERNAL	0.384
Archetype 2: Internal Governance	
SELF_GOV	0.461
OTHER_INTERNAL	0.279
ROUTINE_DEF	0.063
Archetype 3: Historical/Mnemonic Statehood	
SELF_STATE	0.773
COLLECT_MEM	0.746
OTHER_INTERNAL	0.559
ROUTINE_DEF	0.441
HIST_CONT	0.349
Archetype 4: Civic/Moral Statehood (Positive)	
SELF_STATE	0.964
MORAL_AGENCY	0.230
OTHER_INTERNAL	0.186
ROUTINE_DEF	0.132

Note: Only codes with probability > 0.10 shown for clarity.